

MSRE case

zaidabulawi Abulawi

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1 MSRE description

The MSRE was a graphite moderated flowing salt type reactor with a design maximum operating power of 10 MW(th) developed by Oak Ridge National Laboratory (Robertson, 1965). The fuel salt was a mixture of lithium, beryllium, and zirconium fluoride containing uranium or thorium and uranium fluoride. The coolant salt was a mixture of lithium fluoride and beryllium fluoride. The reactor consisted of two flow loops: a primary loop and a secondary loop. The primary loop connected the reactor vessel to a fuel salt centrifugal pump and the shell side of the shell-and-tube heat exchanger. The secondary loop connected the tube-side of the shell-and-tube heat exchanger to a coolant salt centrifugal pump and the tube side of an air-cooled radiator. Two axial blowers supplied cooling air to the radiator. Piping, drain tanks and “freeze valves” made up the remaining components of the heat transport circuits. The heat generated in the core was transferred to the secondary loop through the heat exchanger and ultimately rejected to the atmosphere through the radiator. The current 1-D MSRE model focuses on the primary loop with all the key reactor components represented.

The fuel salt in the MSRE primary loop was $\text{LiF-BeF}_4\text{-ZrF}_4\text{-UF}_4$ according to the design specifications of the MSRE, of which the thermophysical properties are listed in Table 1.

Table 1: Thermophysical properties of the fuel salt.

Property	Unit	$\text{LiF-BeF}_4\text{-ZrF}_4\text{-UF}_4$
Melting temperature, T_{melt}	K	722.15
Density, ρ	kg/m^3	$2553.3 - 0.562 T$
Dynamic viscosity, μ	$\text{Pa}\cdot\text{s}$	$8.4 \times 10^{-5} e^{2390/T}$
Thermal conductivity, k	$\text{W}/(\text{m}\cdot\text{K})$	1.0
Specific heat capacity, c_p	$\text{J}/(\text{kg}\cdot\text{K})$	2009.66

A conventional, cross-baffled, shell-and-tube type heat exchanger was used in the MSRE. The fuel salt flows on the shell side while the coolant salt flows through the tube side. The coolant salt in the heat changer is LiF-BeF_2 (0.66–0.34), of which the major thermophysical properties are summarized in Table 2. Due to the space limitation in the reactor cell, a U-tube configuration is adopted, which results in a heat exchanger of roughly 2.5 m in length. The shell diameter is 0.41 m while the tube has a diameter of 1.27 cm and a thickness of 1.07 mm. Given a triangular arrangement of the tubes, the hydraulic diameters are 2.09 cm (shell-side) and 1.06 cm (tube-side). The construction material of the heat exchanger is Hastelloy[®] N alloy with the properties listed in Table 3. All connecting pipes have a default diameter of 0.127 m. A centrifugal pump is utilized, and its head is adjusted to sustain the flow circulation. The downcomer and lower plenum to the MSRE core are modeled with SAM 1-D components. It is very important to note that the salt equation of state is used for the heat exchanger secondary side.

The thermophysical properties of the fuel salt are defined using three PiecewiseLinear functions. The density function (`fuel_salt_rho_func`) uses two points ranging from 750 K to 1200 K to form a linear fit corresponding to the values used by He (2016). The enthalpy function (`fuel_salt_enthalpy_func`) likewise spans the temperature range of 750–1200 K using two points that approximate the temperature dependence through $c_p T$.

Table 2: Thermophysical properties of the coolant salt in the heat exchanger.

Property	Unit	LiF–BeF ₂ (0.66–0.34)
Melting temperature, T_{melt}	K	728
Density, ρ	kg/m ³	$2146.3 - 0.488 T$
Dynamic viscosity, μ	Pa·s	$1.16 \times 10^{-4} e^{3755/T}$
Thermal conductivity, k	W/(m·K)	1.1
Specific heat capacity, c_p	J/(kg·K)	2390.0

Table 3: Thermophysical properties of Hastelloy[®] N alloy used in the heat exchanger.

Property	Unit	Hastelloy N
Density, ρ	kg/m ³	8860
Thermal conductivity, k	W/(m·K)	23.6
Specific heat capacity, c_p	J/(kg·K)	578

The dynamic viscosity function (`fuel_salt_mu_func`) uses a much finer discretization, consisting of 46 temperature points spanning from 750 K to 1200 K. This provides a detailed representation of the strong temperature dependence of the fuel-salt viscosity. The full set of discretization points and values is listed in Table 6.

Table 4: Fuel salt density function (`fuel_salt_rho_func`).

Temperature (K)	Density (kg/m ³)
750	2285.31
1200	2032.41

Table 5: Fuel salt enthalpy function (`fuel_salt_enthalpy_func`).

Temperature (K)	Enthalpy (J/kg)
750	1.51×10^6
1200	2.41×10^6

Tables 4–6 summarize the temperature-dependent thermophysical properties of the MSRE fuel salt as implemented in the SAM model through user-defined piecewise-linear functions. Table 1 provides the density function `fuel_salt_rho_func`, which linearly interpolates the salt density between 2285.31 kg/m³ at 750 K and 2032.41 kg/m³ at 1200 K, reflecting the expected thermal expansion behavior of molten fluoride fuel salts. Table 2 lists the enthalpy function `fuel_salt_enthalpy_func`, defined as a linear approximation of $c_p T$ over the range 750–1200 K, increasing from 1.51×10^6 J/kg to 2.41×10^6 J/kg. This simplified representation is consistent with common MSRE thermodynamic correlations.

Table 3 contains the detailed viscosity function `fuel_salt_mu_func`, which is specified at 51 discrete temperature points from 750 K to 1200 K. The viscosity decreases monotonically from 2.74×10^{-2} Pa·s at 750 K to approximately 3.13×10^{-3} Pa·s at 1200 K, capturing the strong temperature dependence of molten-salt rheology. These values are linearly interpolated within SAM, providing a smooth and accurate representation of viscosity across the operating temperature range of the MSRE primary loop.

MSRE Primary Loop Topology Description

The SAM model represents the Molten Salt Reactor Experiment (MSRE) primary loop as a closed one-dimensional fluid network beginning in the downcomer, where the fuel salt flows vertically downward toward the inlet plenum. The downcomer is modeled as a straight vertical conduit whose outlet feeds into a junction that directs the salt into the inlet plenum. The inlet plenum is a short horizontal region that distributes the

Table 6: Fuel salt viscosity function (fuel_salt_mu_func).

Temperature (K)	Viscosity (Pa.s)
750	2.7378E-02
760	2.5371E-02
770	2.3557E-02
780	2.1915E-02
790	2.0424E-02
800	1.9069E-02
810	1.7834E-02
820	1.6706E-02
830	1.5674E-02
840	1.4728E-02
850	1.3859E-02
860	1.3060E-02
870	1.2324E-02
880	1.1645E-02
890	1.1017E-02
900	1.0436E-02
910	9.8976E-03
920	9.3976E-03
930	8.9328E-03
940	8.5001E-03
950	8.0969E-03
960	7.7206E-03
970	7.3690E-03
980	7.0402E-03
990	6.7322E-03
1000	6.4434E-03
1010	6.1724E-03
1020	5.9178E-03
1030	5.6783E-03
1040	5.4529E-03
1050	5.2404E-03
1060	5.0400E-03
1070	4.8508E-03
1080	4.6720E-03
1090	4.5029E-03
1100	4.3428E-03
1110	4.1911E-03
1120	4.0473E-03
1130	3.9109E-03
1140	3.7813E-03
1150	3.6582E-03
1160	3.5411E-03
1170	3.4297E-03
1180	3.3235E-03
1190	3.2224E-03
1200	3.1259E-03

incoming fuel salt before it enters the bottom of the reactor core. A branch connects the plenum outlet to the core inlet, ensuring the proper transition between these two components.

The reactor core is modeled as a vertical upward-flow channel containing a volumetric heat source rep-

resenting nuclear heat generation. As the fuel salt flows upward through the core, it absorbs heat and is discharged into the upper plenum. This upper plenum serves as a mixing and expansion region at the core exit and is connected via a branch to a long horizontal pipe section. The fluid then passes through this horizontal pipe, continues through a vertical pipe segment, and finally reaches the pump inlet.

The pump raises the system pressure and drives the fuel salt through the remainder of the primary loop. Downstream of the pump, the flow enters another horizontal pipe and then transitions into the shell side of the heat exchanger. The heat exchanger shell is a long horizontal channel carrying the hot fuel salt while it transfers heat to a secondary coolant flowing through a U-tube bundle. The primary side contains no internal heat source; all heat removal occurs through thermal coupling with the secondary tubes. Two coupled heat structure components represent the tube walls, modeling axial and radial conduction as well as convective heat transfer on both the shell side and tube side.

The secondary coolant enters the first tube leg of the heat exchanger through a velocity boundary condition. It flows horizontally through the first tube, continues through a short vertical bend, and returns through a second long horizontal tube before leaving through a pressure boundary outlet. As it flows, the secondary coolant absorbs heat conducted through the tube walls. The heat structures are fully coupled to the corresponding primary and secondary flow passages, enabling accurate modeling of heat exchange between fluids.

After the fuel salt releases heat in the heat exchanger, it exits the shell-side channel and enters a mixing junction. This junction also receives flow from a short pressure-reference pipe, which is tied to a fixed-pressure boundary condition that anchors the loop's thermodynamic state. The combined flow leaves the junction and travels downward through a vertical pipe segment, then horizontally through another pipe, and finally re-enters the downcomer through the final branch, completing the circulating loop.

In this configuration, the SAM model captures the essential components and flow pathways of the MSRE primary circuit: downward flow through the downcomer, upward flow through the heated core, transit through horizontal and vertical pipes to the pump, heat rejection in the shell side of a U-tube heat exchanger, and the return flow back to the downcomer. This topology enables detailed simulation of steady-state and transient thermal-hydraulic behavior within the MSRE system.

Primary-Loop Fluid Components

Table 7: Components geometry specifications including areas and hydraulic diameters.

Component	Area (m ²)	Dh (m)	Length (m)	Position (x,y,z)	Orientation	n_elems
downcomer	0.1589	0.0508	1.7272	(-0.7366, 1.7272, 0)	(0, -1, 0)	18
inlet plenum	0.3932	0.6997	0.7366	(-0.7366, 0, 0)	(1, 0, 0)	8
core	0.3512	0.6687	1.7272	(0, 0, 0)	(0, 1, 0)	20
upper plenum	0.3932	0.6997	0.4346	(0, 1.7272, 0)	(0, 1, 0)	6
pipe1	0.01267	0.127	1.8288	(0, 2.1618, 0)	(1, 0, 0)	19
pipe2	0.01267	0.127	0.8128	(1.8288, 2.1618, 0)	(0, 1, 0)	9
pipe3	0.01267	0.127	1.0668	(1.8288, 2.9746, 0)	(-1, 0, 0)	11
hx_primarySide	0.10183	0.020945	2.5298	(0.762, 2.9746, 0)	(-1, 0, 0)	26
hx_secondarySide	0.027885	0.010566	-	-	-	-
pipe_ref	0.01267	0.127	0.1	(-1.8678, 2.9746, 0)	(1, 0, 0)	2
pipe4	0.01267	0.127	1.2474	(-1.7678, 2.9746, 0)	(0, -1, 0)	13
pipe5	0.01267	0.127	1.0312	(-1.7678, 1.7272, 0)	(1, 0, 0)	11

Table 7 summarizes the geometric and discretization characteristics of all `PBOneDFluidComponent` objects that make up the MSRE primary loop. Each component is defined by its flow area, hydraulic diameter, axial length, spatial position, flow orientation, and number of finite-volume elements used for discretization. The table reflects the full one-dimensional representation of the primary circuit, beginning with the downcomer, which carries fuel salt downward toward the inlet plenum, followed by the core region where nuclear heat is added. Upstream of the core, the model includes the upper plenum and multiple pipe segments that route the heated fuel salt to the pump and subsequently to the heat exchanger. The heat exchanger consists of both the shell-side passage for the primary fuel salt and the three tube-side segments for the secondary coolant. Additional short pipe sections close the loop, including the reference leg (`pipe_ref`) and the return pipes (`pipe3_s1` and `pipe3_s2`) leading back to the downcomer. Together, these geometric definitions establish the one-dimensional hydraulic network used by SAM to compute mass, momentum, and energy transport throughout the MSRE primary-loop system.

Branch Components

Table 8: Branch components and associated form-loss coefficients.

Branch	Area (m ²)	Loss Coefficients K	Connections (Input → Output)
j_dn_pl	0.1155	(0.0, 0.0)	downcomer(out) → iplnm(in)
j_ip_c	0.1155	(0.0, 0.0)	iplnm(out) → core(in)
j_c_up	0.1155	(0.0, 0.0)	core(out) → uplnm(in)
j_up_ps1	0.1155	(0.0, 0.0)	uplnm(out) → pipe1(in)
j1	0.01292	(0.0, 0.0)	pipe1_s1(out) → pipe2(in)
j2	0.01267	(0.0, 0.0)	pipe3(out) → hx_primary(in)
j3	0.01267	(0.0, 1000, 0.0)	hx_primary(out), pipe_ref(out) → pipe4(in)
j4	0.01267	(0.0, 0.0)	pipe4(out) → pipe5(in)
j5	0.01267	(0.0, 0.0)	pipe5(out) → downcomer(in)

Table 8 summarizes all **PBBranch** elements used to connect the one-dimensional fluid components throughout the MSRE primary loop. Each branch is defined by its hydraulic area, its form-loss coefficients K , and the specific components it links together. These branches represent localized junctions where momentum losses, flow redirection, or geometric transitions occur. For most branches in the MSRE model, the loss coefficients are set to zero, reflecting idealized connections with negligible additional pressure drop. The exception is branch j3, which mixes the primary heat-exchanger outlet flow with the reference leg and includes a large loss coefficient ($K = 1000$) on one of its flow paths to control the distribution and stabilize the loop pressure. The table illustrates how each junction supports the overall hydraulic network: guiding flow from the downcomer into the inlet plenum, from the plenum into the core, from the core to the upper plenum, through downstream piping into the heat exchanger, and finally along the return path back to the downcomer. Collectively, these branch components ensure proper connectivity, flow directionality, and localized momentum modeling within the SAM representation of the MSRE primary circuit.

Heat Exchanger Characteristics

Table 9: Heat exchanger BCs.

Component	Side	Notes
hx_s_in	Boundary	Velocity BC: $v = 1.6$ m/s
hx_s_out	Boundary	Pressure BC: $p = 10^5$ Pa

Table 9 includes the boundary-condition components governing the secondary coolant: **hx_s_in**, which imposes a velocity inlet condition of 1.6 m/s and an inlet temperature of 824.8 K, and **hx_s_out**, which specifies a pressure outlet boundary of 10^5 Pa. Together, these components form a fully coupled representation of the U-tube heat exchanger, capturing the thermal interaction between the primary fuel salt and the secondary Flibe coolant.

Initial and Boundary Conditions

The MSRE primary-loop model is initialized using global conditions that establish a uniform and physically consistent starting state across the fluid and solid regions. The fluid pressure is initialized to 1.0×10^5 Pa, the fluid velocity is set to a small nonzero value of 0.001 m/s, and all fluid and solid temperatures begin at 908.15 K. A small solid temperature scaling factor of 10^{-3} is applied, and gravity is specified in the negative y -direction as $(0, -9.8, 0)$. The pump component is assigned an initial internal pressure of 1.1×10^5 Pa. Within the heat exchanger, the secondary-side tubes (hx_tube1, hx_tube2, hx_tube3) and the coupled tube walls (hx_wall1, hx_wall2) are initialized at a lower temperature of 824.8167 K to represent a colder secondary loop at startup, while the primary-side shell adopts the global initial temperature.

Boundary conditions are imposed on both the secondary loop and a reference location in the primary loop. On the secondary side, the inlet is modeled using a velocity boundary condition (hx_s_in) with $v = 1.6$ m/s and an inlet temperature of 824.8167 K, while the outlet (hx_s_out) is defined using a pressure boundary with $p = 1.0 \times 10^5$ Pa and an outlet temperature of 866.4833 K. On the primary loop, a short pressure-reference leg (pipe_ref) anchors the loop pressure through a boundary condition (ref_p) that fixes the pressure at 1.233351×10^5 Pa and temperature at 908.15 K. This combination of initial and boundary conditions ensures

a stable, well-posed starting solution for the transient or steady-state simulation of the MSRE primary system.